σ-Additive Probability on Orthomodular Lattices: A Survey and an Open Problem

Patrick S. Mahon

Survey note — June 2026

Abstract

The passage from a finitely coherent assignment of probabilities to a countably additive measure is, in the Boolean case, governed by a single condition (countable additivity) and executed by classical machinery (Carathéodory; Loomis–Sikorski; Stone duality). On a *non-distributive* orthomodular lattice (OML) — the algebra of propositions of a quantum system — this machinery breaks, and the question of when finite coherence forces countable behaviour is open. We survey the problem at the level needed to begin work on it. After fixing the apparatus (orthomodular lattices, the gap between orthogonality and meet-zero, the state/meet-additive/charge ladder, the McDonald–Bimbó dual space), we separate the question into two axes. The *extension* axis is classical (Horn–Tarski feasibility) and fully settled: on $L(H)$ no state of any kind extends, by an elementary finite construction; the *descent* axis is genuinely open and rests on a representation that does not exist: a Loomis–Sikorski theorem for σ-complete OMLs. The stake is more than technical: such a representation would furnish a *point-free*, σ-additive probability theory on a non-distributive lattice — the non-Boolean analogue of localic measure theory — built from the entailment relation rather than descended from a sample space. We review what is known and what is ruled out, and close with the two directions in which the open problem might be approached.

1 Introduction

All measurement is finite, yet the frameworks that organise empirical knowledge — probability theory, quantum mechanics — rest on infinite structures. The passage from finite observation to infinite structure is not automatic; it requires a commitment no finite body of evidence can compel. In probability theory the locus of that commitment is *countable additivity*:

$$A_1 \supseteq A_2 \supseteq \cdots, \quad \bigcap_n A_n = \varnothing \quad \Longrightarrow \quad \mu(A_n) \to 0.$$

De Finetti [7] held that only finite additivity is empirically grounded; Kolmogorov [23] adopted σ-additivity as an axiom.

The same tension recurs once observations need not be jointly performable. An observation *context* — a set of measurements performable together — is Boolean; gluing contexts along shared observations (a categorical colimit) returns:

contexts mutually compatible $\Rightarrow$ Boolean algebra, incompatible $\Rightarrow$ non-distributive OML

(Gunji et al. [16]; the dichotomy organises the companion paper [24]). The OML is *forced* by the context structure, not postulated. Incompatibility — quantum complementarity, but equally contextual experimental design or interfering sensors — drives the algebra out of the Boolean world. One caveat fixes the scope: for non-quantum examples the incompatibility must be *operationally imposed* (contexts genuinely non-co-realizable, no joint probability space); it cannot arise intrinsically from one classical system's dynamics, since any finite family of classical observables on a common space jointly distributes (Kolmogorov), hence stays Boolean — contextuality there is failure of a global section (Abramsky–Brandenburger [1]; Fine [11]). The closed subspaces of a Hilbert space H form the prototype $L(H)$, not the premise.

On a non-distributive OML, Carathéodory's outer measure — needing a Boolean algebra of sets — does not apply, and the rescue is local to $L(H)$:

- For $L(H)$, Gleason's rigidity theorem [14] rescues it; Hilbert space is special and no general-OML analogue is known.
- Non-distributivity *per se* does not remove the 2-valued homomorphisms the Boolean engine needs — MO_3 is non-distributive yet carries them (§2.1, §4) — but it removes the *guarantee*; for $L(H)$, $\dim \geq 3$ they fail outright (Kochen–Specker [22]).

Whether enough survive is a question of concreteness, not non-distributivity — the qualifier on which the open problem turns (§4, §5).

The organising split. This note asks the shared question — *when does finite coherence force countable behaviour?* — on an OML. The vague question splits into two axes that the Boolean case fuses, and the survey is built around the split:

Axis	**Asks**	**Status**
Extension	does a state extend to a charge on $S_0(A)$?	settled (on $L(H)$, *none* does)
Descent	does the charge concentrate on physical points?	**open**

The relational standpoint. Two established moves motivate the approach:

- De Finetti [7] — probability without a presupposed sample space: coherent commitments about events, not measures on a given Ω.
- Birkhoff–von Neumann — propositions as a lattice (the OML of §2), not as subsets of a phase space.

Combining them, we start from the propositions and their entailment order and ask what that relational structure alone determines — the measure-theoretic counterpart of pointless (localic) topology. The dual is then not a space of points but the McDonald–Bimbó space of *filters* (§2.3), and a positive answer to the open problem would be a *point-free* σ-additive probability theory on a non-distributive lattice — the non-Boolean analogue of localic measure theory (§5; developed further in [25]).

2 Preliminaries

We collect the lattice-theoretic apparatus; all notions are standard.

2.1 Orthomodular lattices, and the orthogonal/meet-zero gap

Definition 2.1 (Orthocomplemented lattice)**.** A *bounded lattice* is a partially ordered set in which every pair a, b has a join $a \vee b$ and meet $a \wedge b$, with greatest element $\mathbf{1}$ and least element $\mathbf{0}$. An *orthocomplementation* is a map $a \mapsto a^\perp$ satisfying, for all a, b:

(i) $a \wedge a^\perp = \mathbf{0}$ and $a \vee a^\perp = \mathbf{1}$ (complement),
(ii) $a^{\perp\perp} = a$ (involution),
(iii) $a \leq b \Rightarrow b^\perp \leq a^\perp$ (order-reversing).

Definition 2.2 (Incomparable; meet-zero; orthogonal)**.** For elements a, b of an orthocomplemented lattice, three relations of increasing strength:

- *incomparable*: $a \not\leq b$ and $b \not\leq a$ — neither entails the other (an order condition);

- *meet-zero*: $a \wedge b = \mathbf{0}$ — no common refinement;
- *orthogonal*, written $a \perp b$: $a \leq b^{\perp}$ (equivalently $b \leq a^{\perp}$) — one entails the other's negation, *decisively incompatible.*

Each implies the one above it ($a \perp b \Rightarrow a \wedge b = \mathbf{0} \Rightarrow$ incomparable, for $a, b \notin \{\mathbf{0}, \mathbf{1}\}$). For two *distinct atoms*, incomparable and meet-zero coincide — an atom has only $\mathbf{0}$ below it, so $a \wedge b \in \{\mathbf{0}, a\}$, and $a \wedge b = a$ would force $a \leq b$; orthogonality remains strictly stronger.

> **The gap.** The implication $a \perp b \Rightarrow a \wedge b = \mathbf{0}$ *reverses* in every Boolean algebra but **fails** once the lattice is non-distributive. *Meet-zero is then strictly weaker than orthogonal* — the distinctions developed below all turn on this.

Definition 2.3 (Orthomodular lattice)**.** An orthocomplemented lattice A is *orthomodular* if

$$a \leq b \quad \Longrightarrow \quad b = a \vee (a^{\perp} \wedge b). \tag{1}$$

A *Boolean algebra* is the special case where $\wedge$ distributes over $\vee$; every Boolean algebra is orthomodular, not conversely. The motivating non-Boolean example is $L(H)$, the closed subspaces of a Hilbert space with $a^{\perp}$ the orthogonal complement.

Definition 2.4 (Completeness notions)**.** Two countable-completeness conditions on an OML A are distinguished throughout, the second strictly weaker. Here $\bigvee_n a_n$ denotes the *join* (least upper bound) of $\{a_n\}$ in the order of A, unique when it exists; "$\exists \bigvee_n a_n \in A$" asserts that it does.

- A is *σ-complete* if
$$\forall \{a_n\}_{n\in\mathbb{N}} \subseteq A : \quad \exists \textstyle\bigvee_n a_n \in A$$
(equivalently every countable family has a meet, by $\bigwedge_n a_n = (\bigvee_n a_n^{\perp})^{\perp}$);
- A is *σ-orthocomplete* if
$$\forall \{a_n\}_{n\in\mathbb{N}} \subseteq A \ \big(a_i \perp a_j \text{ for } i \neq j\big) : \quad \exists \textstyle\bigvee_n a_n \in A.$$

The orthogonal families are a subclass of all countable families, so σ-completeness quantifies over a strictly larger domain and is the stronger hypothesis:

$$\{a_n\} \text{ pairwise orthogonal } \subseteq \{a_n\} \text{ arbitrary} \qquad \Longrightarrow \qquad \sigma\text{-complete} \Rightarrow \sigma\text{-orthocomplete}.$$

Only the weaker condition is needed below. For a state s, σ-additivity is the requirement

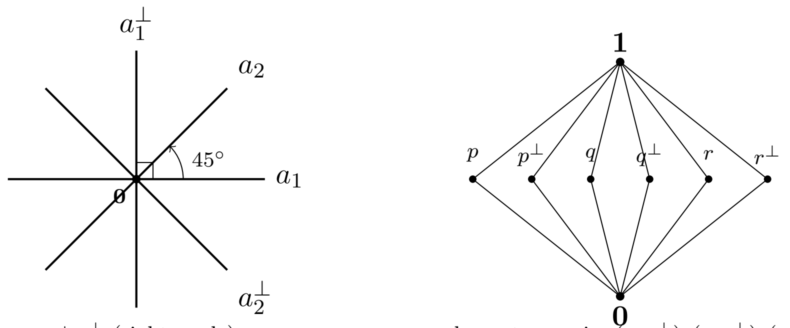
$$s\Big(\textstyle\bigvee_n a_n\Big) = \sum_n s(a_n) \qquad \text{for every countable } \{a_n\} \text{ with } a_i \perp a_j \ (i \neq j),$$

whose left side invokes $\bigvee_n a_n$ *only* for orthogonal families — so σ-orthocompleteness is exactly the hypothesis that makes it well-posed. Full σ-completeness is the stronger condition manipulated by the Loomis–Sikorski representation (§2.4).

Example 2.5 (MO_n, and the gap made concrete)**.** MO_n is the lattice with $\mathbf{0}$, $\mathbf{1}$, and n complementary pairs of incomparable atoms, all sharing only $\mathbf{0}$ and $\mathbf{1}$. The geometric model is n lines through the origin: each line a is orthogonal to its perpendicular $a^{\perp}$, while two atoms from distinct pairs are non-perpendicular lines. It is orthomodular and non-distributive ($n \geq 2$): two atoms drawn from *distinct* complementary pairs have

$$a \wedge b = \mathbf{0} \quad \text{(meet only at the origin)} \qquad \text{yet} \qquad a \not\leq b^{\perp},$$

so meet-zero $\neq$ orthogonal already here. MO_3 is the smallest non-distributive OML.



$a_1 \perp a_1^\perp$ (right angle), $a_1 \not\perp a_2$ but $a_1 \wedge a_2 = \mathbf{0}$

complementary pairs $(p, p^\perp), (q, q^\perp), (r, r^\perp)$; any two non-paired atoms meet at $\mathbf{0}$, join at $\mathbf{1}$

Figure 1: MO_n illustrated. *Left:* the geometric model for MO_2 — four atoms as lines through the origin in $\mathbb{R}^2$. The pair $a_1, a_1^\perp$ meets at a right angle (orthogonal), whereas a_1, a_2 meet at 45°: their lattice meet is still $\mathbf{0}$ (only the origin in common), yet they are *not* orthogonal — the meet-zero $\neq$ orthogonal gap. *Right:* the Hasse diagram of MO_3, the smallest non-distributive OML: three complementary pairs of incomparable atoms between $\mathbf{0}$ and $\mathbf{1}$.

Definition 2.6 (OMP; order-determining states; concrete logic)**.** An *orthomodular poset* (OMP) weakens Definition 2.3: joins $a \vee b$ are required only for *orthogonal* pairs. A set S of states on A is *order-determining* if it separates the order, i.e.

$$\big(s(a) \le s(b) \text{ for all } s \in S\big) \implies a \le b.$$

An OMP is a *concrete logic* (equivalently *set-representable*) if it embeds into $(\mathcal{P}(X), \subseteq, X \setminus (-))$ for some set X, preserving existing orthogonal joins — i.e. its propositions are modelled as actual *sets of outcomes*. Concreteness is equivalent to admitting an order-determining set of *two-valued* states (Gudder [15]), and it is the property that separates the open frontier from the settled cases (§5).

Remark 2.7 (Concreteness does not close the gap)*.* A set-representable OMP needs only closure under complement and *disjoint* union [3]; intersection-closure would force a field of sets, hence Boolean. The lattice meet $a \wedge b$ is the greatest L-member inside $A \cap B$, so

$$\underbrace{A \cap B = \varnothing}_{\text{orthogonal}} \implies \underbrace{a \wedge b = \mathbf{0}}_{\text{meet-zero}}, \qquad \text{but not conversely.}$$

Concreteness does not force the converse: MO_3 is itself concrete (Definition 2.6; Gudder [15]), so the paradigmatic meet-zero-$\neq$-orthogonal example is already a concrete logic. What closes the gap is a richness/atomicity condition, not concreteness:

$$a \wedge b = \mathbf{0} \Rightarrow a \perp b \quad \iff \quad \text{every nonempty } A \cap B \text{ contains a nonzero } L\text{-member},$$

the abstract form being Tkadlec's *Boolean orthoposet* condition [34] (which does *not* mean Boolean *algebra*).

2.2 The state/meet-additive/charge ladder

Definition 2.8 (State; meet-additive state; charge-extendible)**.** Let A be an OML. A *state* is a map $s : A \to [0, 1]$ with $s(\mathbf{1}) = 1$ that is additive on *orthogonal* pairs: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$; that is, a probability assignment whose additivity is guaranteed only on decisively incompatible propositions. Three strengthenings, differing in which pairs additivity is demanded on and strictly increasing in general (Example 2.9 on MO_3):

(1) *State*: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$.
(2) *Meet-additive*: $s(a \vee b) = s(a) + s(b)$ whenever $a \wedge b = \mathbf{0}$ (binary, strictly stronger — it reaches the gap pairs of Definition 2.2). We avoid the word "valuation" here: in lattice theory it denotes the modular function $v(a)+v(b) = v(a \wedge b)+v(a \vee b)$, a different condition.
(3) *Charge-extendible*: $\sum_i s(a_i) \leq 1$ for every pairwise-meet-zero family $\{a_i\}$ ($a_i \wedge a_j = \mathbf{0}$, $i \neq j$; strictly stronger again). This is the condition necessary for extension to a charge on the dual space (Definition 2.12) and the *extension* condition of §3.

Example 2.9 (The ladder is strict)**.** On MO_3 with complementary pairs $(p, p^\perp), (q, q^\perp), (r, r^\perp)$, the maximally mixed $s \equiv \frac{1}{2}$ is meet-additive (clears (2)) yet fails (3) at $\{p, q, r\}$: $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} > 1$. So no state on MO_3 is charge-extendible.

Definition 2.10 (Normal and singular states on $L(H)$)**.** On $A = L(H)$ a state s is

- *normal* if it is completely additive on every orthogonal family of projections,

$$s\Big(\bigvee_{i \in I} p_i\Big) = \sum_{i \in I} s(p_i) \quad \text{for all (not only countable) orthogonal } \{p_i\},$$

  equivalently $s(\cdot) = \mathrm{tr}(\rho\, \cdot)$ for a density operator ρ (Gleason [14], dim$\geq$3);
- *singular* (purely) if

$$s(e) = 0 \quad \text{for every finite-rank projection } e.$$

Every state decomposes into a normal and a singular part (Takesaki); the two classes are exactly those distinguished by the arguments of §3.

2.3 The McDonald–Bimbó dual space

Definition 2.11 (Dual space; representation map)**.** The McDonald–Bimbó duality [26] assigns to an OML A a compact space

$$S_0(A) = \big(F(A), \subseteq, \perp_A, P(A), T\big),$$

where $F(A)$ is the set of *filters* of A, ordered by $\subseteq$ and carrying an orthogonality relation $\perp_A$ and a Stone topology T, and $P(A) = \{\, \uparrow p : p \in A \,\} \subseteq F(A)$, with $\uparrow p = \{a \in A : a \geq p\}$, is the set of *principal* filters — the "physical points," the realisation datum (canonical, unlike the non-canonical pure points of the Boolean case). For $a \in A$ set $h(a) = \{x \in F(A) : a \in x\}$; then h is an OML isomorphism onto the $\perp$-stable clopens $\mathrm{CO}(S_0(A))^\dagger$. Define $\mu(h(a)) = s(a)$. Here $h(a)$ is the set of points at which a holds and μ is the state transported onto those sets; the open problem is whether μ is preserved under countable operations on them.

Definition 2.12 (Charge)**.** A *charge* on $S_0(A)$ is a map $\nu : \mathrm{CO}(S_0(A)) \to [0, 1]$ on the *full* Boolean algebra of clopen subsets (not merely the $\perp$-stable ones in the image of h), with

$$\nu(S_0(A)) = 1, \qquad \nu(U \cup V) = \nu(U) + \nu(V) \ \text{ whenever } U \cap V = \varnothing.$$

A state s *extends to a charge* if some such ν satisfies $\nu(h(a)) = s(a)$ for all $a \in A$. Because h preserves meets, a pairwise-meet-zero family in A maps to disjoint clopens, so a charge necessarily satisfies $\sum_i s(a_i) \leq 1$ over such families — the charge-extendibility condition of Definition 2.8(3).

Remark 2.13 (The duality is finitary)*.* The representation map h satisfies

$$h\Big(\bigvee_{i \leq n} a_i\Big) = \Big(\bigcup_{i \leq n} h(a_i)\Big)^{\perp\perp} \ \text{(finite)}, \qquad h\Big(\bigvee_{n} a_n\Big) \neq \Big(\bigcup_{n} h(a_n)\Big)^{\perp\perp} \ \text{(countable)}:$$

finitary OML homomorphisms need not preserve infinite suprema. This is the central obstruction the open problem of §5 confronts, and it is self-dual: since $a \mapsto a^\perp$ is an order anti-isomorphism, $\bigwedge_n a_n = (\bigvee_n a_n^\perp)^\perp$, so the failure for countable meets is the De Morgan dual of the failure for countable joins. The rival duality of Cannon–Döring [4] is also finitary and carries no measure; we use McDonald–Bimbó because it carries $P(A)$ as explicit structure.

2.4 The Boolean engine: Stone duality

In the Boolean case the descent equivalence

$$\sigma\text{-additive} \iff \text{the measure concentrates on the physical points}$$

is proved through *Stone duality*, in four steps [25]:

(S1) a Boolean event algebra B embeds in the clopen algebra $\mathrm{CO}(\mathrm{St}\,B)$ of its Stone space;
(S2) a charge on B extends to a Baire measure on $\mathrm{St}\,B$ (Carathéodory);
(S3) σ-additivity $\iff$ the measure vanishes on meagre Baire sets (Rao–Rao [32]);
(S4) meagre-null $\iff$ the measure lives on the ultrafilters closed under countable intersection = the realised points.

Steps (S3)–(S4) are the measure-theoretic face of the *Loomis–Sikorski* representation,

$$\sigma\text{-complete Boolean algebra} \;\cong\; (\sigma\text{-tribe of sets}) \,/\, (\sigma\text{-ideal}),$$

the countable-join bookkeeping that drives the equivalence. The contrast with the OML case is exact:

Boolean	**OML**
Stone duality + Loomis–Sikorski (the engine)	**no analogue** (§4.3)

Supplying such an engine, or proving none exists, is the open problem of §5.

Remark 2.14 (Why the bridge breaks: meet-closed but not join-prime). A “realised” point x carries two conditions, which Rao–Rao’s “vanishes on meagre sets” makes coincide in the Boolean case:

(M) closed under countable meets; (J) $x \ni \bigvee_n a_n \Rightarrow x \ni a_n$ some n (off the join-defect).

Non-distributivity separates them. For a principal filter $x = \uparrow p$:

(M) always holds.
(J) can fail: $p \le \bigvee_n a_n$ does not force $p \le a_n$.

Witness in $L(H)$. $a_n = \mathrm{span}(e_n)$, $p = \mathrm{span}(e_1 + e_2)$:

$$p \;\le\; \textstyle\bigvee_n a_n = H, \qquad \text{yet} \qquad p \;\not\le\; a_n \;\text{ for every } n.$$

So $\uparrow p$ is a realised point lying inside the join-defect $D = h(\bigvee_n a_n) \setminus \bigcup_n h(a_n)$. Topologically $\uparrow p$ is *isolated* in $S_0(L(H))$, hence:

$$D \text{ non-meagre (OML)} \qquad \text{vs.} \qquad D \text{ nowhere dense (Boolean).}$$

This is the finitary obstruction of Remark 2.13 on the dual: the category route (clopens modulo the meagre ideal) fails on the same D as the measure route, and reduces to the MacNeille completion of §4.3(ii).

3 Two axes: extension and descent

The central question — *when does a state s on an OML A extend to a σ-additive measure on $S_0(A)$?* — splits into two **independent** axes that the Boolean case fuses. A state s on A faces:

	Extension axis	**Descent axis**
Question	Does μ extend to a finitely additive charge on the *full* Boolean clopen algebra of $S_0(A)$?	Once extended, does the measure concentrate on the physical points $P(A)$?
Governed by	the meet-zero/orthogonal gap (§2.1)	σ-additivity — needs the missing engine (§2.4)
Status	**settled**: Horn–Tarski / Pitowsky feasibility; finite case = decidable LP; can fail for every state.	**open**. *The descent residue.*

Two observations pin down the extension axis.

Proposition 3.1 (Finite case). *For finite A the extension condition is a decidable linear program (Horn–Tarski [20] / Pitowsky correlation-polytope feasibility [29]); since h preserves meets, meet-zero elements map to disjoint clopens and a charge must satisfy $\sum_i s(a_i) \le 1$ over pairwise-meet-zero families. It can fail for* every *state (*MO_3*, Example 2.9).*

Proposition 3.2 (Infinite $L(H)$, all states). *Let* $\dim H = \infty$*.* No *state on $L(H)$ — normal or singular — extends to a charge on $S_0(L(H))$.*

Proof. The argument embeds a copy of MO_2 in $L(H)$ whose four atoms are infinite-dimensional, so that the obstruction reaches the singular states as well. Write $H = H_0 \otimes \mathbb{C}^2$ with H_0 infinite-dimensional, and fix an orthonormal basis e, f of $\mathbb{C}^2$. Set

$$a_1 = H_0 \otimes \mathbb{C}e, \quad a_1^\perp = H_0 \otimes \mathbb{C}f, \quad a_2 = H_0 \otimes \mathbb{C}(e+f), \quad a_2^\perp = H_0 \otimes \mathbb{C}(e-f).$$

These four infinite-dimensional subspaces form a copy of MO_2 in $L(H)$:

$$a_1 \perp a_1^\perp, \quad a_2 \perp a_2^\perp, \quad a_i \vee a_i^\perp = H; \qquad a_i \wedge a_j = \mathbf{0} \text{ but } a_i \not\perp a_j \text{ (cross pairs).}$$

Orthoadditivity with $s(\mathbf{1}) = 1$ gives $s(a_i) + s(a_i^\perp) = 1$ for $i = 1, 2$, using only additivity on orthogonal pairs, so it holds for *every* state. The family $\{a_1, a_1^\perp, a_2, a_2^\perp\}$ is pairwise meet-zero, and since h preserves meets ($h(a \wedge b) = h(a) \cap h(b)$), its members map to disjoint clopens; hence any charge satisfies $\sum_i s(a_i) \le 1$. The two counts collide:

$$\sum_i s(a_i) = 2 \text{ (orthoadditivity)} \qquad \text{versus} \qquad \sum_i s(a_i) \le 1 \text{ (any charge).}$$

The construction is finite ($n = 4$); σ-completeness of the lattice plays no role. □

The MO_2 count is standard (orthoadditivity together with $s(\mathbf{1}) = 1$; see [21]) and $\sum s(a_i) = 2$ is already state-independent in $L(\mathbb{C}^2)$. The role of the *infinite-dimensional* realisation is to extend the obstruction to the singular states ($s(e) = 0$ on every finite-rank e):

Atoms of MO_2	$s(a_i) + s(a_i^\perp)$	**Reaches**
lines in $L(\mathbb{C}^2)$ (finite rank)	$= 1$ only if s charges finite rank	normal states only
$H_0 \otimes (\cdot)$, $\dim H_0 = \infty$	$= 1$ regardless of finite-rank behaviour	**all states, incl. singular**

Keeping each atom infinite-dimensional drops the finite-rank precondition of the line argument (Remark 3.3), and that is what closes the extension axis on $L(H)$ completely.

Remark 3.3 (An alternative argument for normal states only). A second, technique-disjoint argument reaches the normal states without the MO_2 family. If $s(e) > 0$ on some finite-dimensional e, it is positive on a 2-plane e_0; taking k pairwise-non-orthogonal lines $p_1, \ldots, p_k \subset e_0$ yields $2k$ pairwise-meet-zero lines with $s(p_i) + s(p_i^\perp) = s(e_0)$, so a charge forces

$$k\, s(e_0) \leq 1 \quad \text{for all } k \quad \Longrightarrow \quad s(e_0) = 0,$$

contradicting $s(e_0) > 0$. Lifting s to a functional via Mackey–Gleason / Bunce–Wright [2] ($B(H)$ has no type I_2 summand) and applying the Takesaki normal/singular decomposition identifies which states the argument reaches:

reached	nonzero normal part ($s(e) > 0$ on some finite e); includes every normal (Gleason) state
missed	purely singular ($s(e) = 0$ for all finite e); e.g. ultrafilter vector states, Calkin-algebra pullbacks

Proposition 3.2 subsumes the missed case; this argument is recorded because its technique is independent and locates the singular states precisely.

4 What is known

The extension axis is settled — on $L(H)$ no state extends (Proposition 3.2); the descent axis turns on whether the Boolean engine of §2.4 has an OML analogue. Three things bear on that:

1. the Boolean benchmark such an analogue must match (§4.1);
2. the partial OML results, positive and obstructive (§4.2);
3. the routes already known to be blocked (§4.3).

4.1 The Boolean benchmark

The conditions an OML engine would have to reproduce. The Kelley–Vladimirov–Pták characterisation (Fremlin [12], Thm 391D) is complete: a Boolean algebra carries a strictly positive σ-additive measure iff it satisfies all three conditions below — each of which loses its meaning off the distributive world.

Boolean (KVP) condition	OML analogue
Dedekind σ-completeness	exists (σ-orthocompleteness)
weak (σ, ∞)-distributivity	**none** — relies on $\bigwedge$ distributing over $\bigvee$
chargeability	**none** — no structural characterisation on OMLs

4.2 Positive and obstructive OML results

Positive (require structure rich enough to approximate Hilbert-space geometry):

- Gleason [14] — $L(H)$, $\dim \geq 3$; σ-additivity *on the lattice*.
- Bunce–Wright [2] — JBW projection lattices; Chetcuti–Dvurečenskij [5].

Obstructive (Boolean machinery cannot be transplanted):

- Pták–Pulmannová [31] — a *finitary* condition (*subadditive*-unitality $s(a \vee b) \leq s(a) + s(b)$, no countable joins) already collapses the *lattice* to Boolean (their Thm 1); the OM*poset* version fails (Müller's set-representable counterexample [27]).
- Navara [28] — regularity does not force σ-additivity. Pták [30] — exotic state spaces via Greechie pasting.

4.3 Three blocked routes to a σ-OML engine

The descent argument needs an OML analogue of the Boolean engine — a σ-Stone duality, or equivalently a Loomis–Sikorski representation. The three known routes are all blocked.

(i) **RDP / effect-algebra.** Loomis–Sikorski holds for σ-MV algebras [9] and monotone σ-effect algebras *with* the Riesz Decomposition Property [10]; but a lattice effect algebra has RDP iff it is MV, and OML $\cap$ MV = Boolean (Kalmbach [21]), so the hypothesis that would yield the representation collapses the class:

$$\text{OML} + \text{RDP} \iff \text{Boolean}.$$

(ii) **MacNeille completion.** The MacNeille completion of an OML need not be orthomodular (Harding [17]); one cannot complete to absorb countable joins. The topological version of this route — clopens of $S_0(A)$ modulo the meagre ideal — is this same completion in another form (the regular-open algebra), and collapses non-distributivity (Remark 2.14).

(iii) **σ-Stone duality.** None exists. McDonald–Bimbó is finitary (Remark 2.13); Freytes' theory [13] of σ-complete OMLs is equational, with no set/tribe representation.

4.4 Prior art on point-free quantum probability

Every existing point-free or directed-context construction fails at least one of the two requirements — σ-additivity and a natively non-distributive structure — as tabulated below.

Gleason tradition
Varadarajan [35], Kalmbach [21], Pták–Pulmannová [31]: σ-additive on a non-distributive OML, but real-valued on a fixed lattice — not point-free.

Döring 2009 [8] point-free on the spectral presheaf over a directed poset — but provably only *finitely additive*, on a distributive Heyting algebra.

Bohrification (HLS)
[18, 19] point-free and the internal valuation *is* σ-additive (Scott-continuous) — but factors through an internally *distributive* locale, σ-additive only per commutative context. Non-distributivity is tamed, not carried natively.

Localic valuations
Vickers [36], Simpson [33], Coquand–Spitters [6]: point-free σ-additive, but on *frames* — distributive by definition.

OML Stone dualities
McDonald–Bimbó [26] and Cannon–Döring [4]: purely structural, no measure, no σ-version.

Each entry satisfies some of the requirements but not all. The open case is their conjunction:

The descent residue (unoccupied): point-free $\times$ σ-additive $\times$ natively non-distributive $\times$ directed-system-built.

(The directed-context-yields-non-distributivity construction is not new — it is the shape of Bohrification and of colimit-over-Boolean-contexts generation. Novelty would lie only in the σ-additive $\times$ point-free $\times$ native combination.)

Remark 4.1 (The boundary of the negative results). Two qualifications delimit what is ruled out.

(i) **No impossibility theorem covers the concrete class.** The known no-go results are finite or two-valued, and the one positive occupant is non-concrete:

$$L(H)\ (\dim\geq 3): \quad \text{no two-valued states at all (Kochen–Specker [22]),}$$

a fortiori none order-determining. Concreteness is the qualifier on which the open frontier turns.

(ii) **No σ-Loomis–Sikorski exists**, and this is visible in the dualities: both take infinite joins as a closure, not a union,

$$\text{Cannon–Döring: } \mathrm{cls}(\textstyle\bigcup S_i); \qquad \text{McDonald–Bimbó: } (\textstyle\bigcup h(a_n))^{\perp\perp}.$$

This is not to be overstated: Gudder's concrete logics [15] *are* set-representable non-Boolean orthoposets — but as point-ful posets, not via a σ-Loomis–Sikorski theorem.

5 The open problem

Open Problem. Does a σ-complete, concrete, non-Boolean, infinite OML admit a Loomis–Sikorski-type representation (a σ-tribe of sets modulo a σ-ideal), or a countable-join-preserving σ-Stone duality — or can one prove that none exists?

Remark 5.1 (Whether the class is inhabited). A member is *descent-relevant* only if it carries an infinite pairwise-orthogonal family (so σ-additivity exceeds finite additivity) *and* meet-zero $\neq$ orthogonal pairs. The completeness at issue is therefore σ-orthocompleteness, not σ-completeness (Definition 2.4).

The discriminating condition (element-versus-family; within one orthogonal family it is vacuous, as such a family is Boolean and there meet-zero $=$ orthogonality):

$$(\bigstar) \quad \exists\, p,\ \exists\,\{a_n\} \text{ infinite orthogonal}, \quad p\wedge a_n = 0\ (\forall n), \quad p \not\perp a_n\ (\forall n). \tag{2}$$

Candidate	Status vs. the four hypotheses	Concrete?	$(\bigstar)$?
MO_κ (κ inf.)	max orthogonal family $=$ 2; descent vacuous; carries MO_2 obstruction	yes	**no** (vacuous)
$L(H)$, $\dim\geq 3$	skew ray P_f vs. basis $\{P_{e_n}\}$, $f=\sum c_n e_n$, $c_n\neq 0$	**no** (KS)	**yes**
Navara $\mathcal{L}$ [28]	infinite, σ-orthocomplete, rich — but stateless block	**no**	yes
$\mathcal{L}_2$ (substitution)	Navara's $\mathcal{L}$ with block := MO_2	**yes**	**open**

The last row is the open case. Write $\mathcal{L}_2$ for Navara's construction with the stateless block replaced by MO_2. It is concrete (sub-OMPs and products of concrete logics are concrete) and σ-orthocomplete: Navara's closure step [28, p. 428] is independent of the states on the block, and the one step that uses the block — "$f_j(m)=0$ and $f_i\perp f_j$ imply $f_j=0$ on the block" — holds in MO_2, where the only elements $\leq a^\perp$ are $\{0,a^\perp\}$. Hence the class of the Open Problem is inhabited, and the remaining question is whether $\mathcal{L}_2$ satisfies (2).

A candidate witness. Let the atoms of MO_2 be $a, a^\perp, b, b^\perp$ with $a \wedge b = 0$ and $a \not\leq b$. Navara's elements have the form $\bigvee_{C \in \mathcal{F}} v_C \,|\, C$ with $\mathcal{F}$ mutually disjoint, and MO_2 caps within-block orthogonal families at 2, so an infinite orthogonal family requires infinitely many disjoint C_n. Set

$$a_n := a \,|\, C_n, \qquad p := \bigvee_n b \,|\, C_n \quad (p \in \mathcal{L}_2 \text{ by } \sigma\text{-orthocompleteness}). \tag{3}$$

The constancy condition is imposed coordinatewise and the C_n are disjoint, so p is admissible; and in $\mathcal{W}$ one has $p \wedge a_n = 0$ (on C_n, $a \wedge b = 0$; off C_n, disjoint supports) and $p \not\leq a_n$ (on C_n, $a \not\leq b$). Thus (2) holds in $\mathcal{W}$.

The remaining step. Navara notes that the lattice operations of $\mathcal{L}$ need not coincide with those of $\mathcal{W}$, and orthogonality is defined through the orthocomplement, so it remains to determine whether (2) transfers from $\mathcal{W}$ to $\mathcal{L}_2$. The question reduces to a single relation:

$$\boxed{(\dagger)\quad \text{In } \mathcal{L}_2, \quad \text{is} \quad a \,|\, C \not\leq b \,|\, C\,? \quad (a, b \text{ distinct non-complementary } \mathrm{MO}_2 \text{ atoms})} \tag{4}$$

$(\dagger)$ holds $\Rightarrow$ p witnesses (2); the conjecture fails, and $\mathcal{L}_2$ realises
$\Rightarrow$ the relational-probability structure of §5.
$(\dagger)$ fails $\Rightarrow$ $\mathcal{L}_2$ segregates, and $L(H)$ remains the only known instance.

Two qualifications: the relevant completeness is σ-orthocompleteness rather than σ-completeness; and $L(H)$ and Navara's construction fail concreteness for distinct reasons (the Kochen–Specker colouring obstruction versus an imported stateless block).

Either resolution is consequential:

Positive a relational probability theory without realisations — σ-additive probability from the entailment relation of a non-distributive OML, the non-Boolean analogue of localic measure theory.

Negative an impossibility theorem closing the entire cluster, both residues, at once.

The art constrains it from both sides: *no* impossibility theorem covers the concrete class (Remark 4.1), yet *all three* constructive routes are blocked (§4.3). What is required is not a new idea in the abstract but a representation that bypasses all three routes.

Remark 5.2 (Why $L(H)$+Gleason does *not* already settle this — the (A)/(B) point-space equivocation). "Probability from relations, point-free" may *look* delivered by $L(H)$+Gleason. It is not: the appearance rests on conflating two point-spaces, and *realisation* is defined as concentration on (B).

	Point-space	**Gleason's relation to it**
(A)	Hilbert rays of H	*requires* them — ρ in $s = \mathrm{tr}(\rho\,\cdot)$ is built from rays
(B)	MB dual filters $P(A)$	*removes* them, but only by being a representation theorem

Gleason is thus the *most* point-ful result available (every lattice state $\mapsto$ the point datum ρ). What $L(H)$ exhibits is a *separation*, not point-freeness:

on the lattice	σ-additive measures exist (Gleason, normal states);
on the dual $S_0(L(H))$	no state extends to a charge (Prop. 3.2),
	so a fortiori none concentrates on $P(A)$.

It cannot serve as the existence proof required, so the σ-OML representation of §5 is the only candidate engine.

6 Directions

With the extension axis closed on $L(H)$ (Proposition 3.2), a single direction remains.

The general construction

Construct a Loomis–Sikorski-type representation, or a countable-join-preserving σ-Stone duality, for a concrete non-Boolean infinite σ-OML — or prove that none exists. This resolves the open problem directly, but no partial construction is currently in hand. This is the *descent*-axis residue: the obstruction is the failure of h to preserve countable joins (Remark 2.13), not the orthogonal/meet-zero gap, and so Proposition 3.2 does not reach it.

A second candidate direction — the singular case for $L(H)$ — is now closed. One might have hoped a singular state would extend where no normal state does, providing the first explicit "extends but may fail to concentrate" example that the general construction lacks; Proposition 3.2 rules this out, catching every state, singular included. The reduction is recorded because the singular case was expected to require infinitary methods and does not (Remark 6.1).

Remark 6.1 (Finite versus countable: finite suffices). Is the obstruction to extending a singular state realised by a *finite* or only by a *countable* pairwise-meet-zero family? The two arguments divide on exactly this point:

line argument (Rmk 3.3)	finitary, but **vacuous** on singular ($s(e) = 0$ $\forall$ finite e)
MO_2**, ∞-dim atoms (Prop. 3.2)**	finite ($n = 4$), $\sum_i s(a_i) = 2$ state-independently

So the singular obstruction is *not* entangled with the descent-axis passage, as the vacuity of the line argument might suggest: it is realised by a finite family, hence a self-contained finitary problem, separable from the general construction and not governed by the finitary-to-σ passage.

References

[1] Samson Abramsky and Adam Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13:113036, 2011.

[2] Leslie J. Bunce and J. D. Maitland Wright. The Mackey–Gleason problem. *Bulletin of the American Mathematical Society*, 26(2):288–293, 1992.

[3] Dominika Burešová and Pavel Pták. On the set-representable orthomodular posets that are point-distinguishing. *International Journal of Theoretical Physics*, 62, 2023. Also arXiv:2401.13798.

[4] Sarah Cannon. *The Spectral Presheaf of an Orthomodular Lattice: A Generalisation of Stone Duality to Orthomodular Lattices*. PhD thesis, University of Oxford, 2013. See also Cannon and Döring, "A generalisation of Stone duality to orthomodular lattices," in *Contemporary Logic and Computing* (2018).

[5] Emmanuel Chetcuti and Anatolij Dvurečenskij. On the Gleason theorem for effect algebras. *Mathematica Slovaca*, 55(3):297–308, 2005.

[6] Thierry Coquand and Bas Spitters. Integrals and valuations. *Journal of Logic & Analysis*, 1(3):1–22, 2009.

[7] Bruno de Finetti. *Theory of Probability*, volume 1. Wiley, 1974.

[8] Andreas Döring. Quantum states and measures on the spectral presheaf. *Advanced Science Letters*, 2(3):291–301, 2009. arXiv:0809.4847.

[9] Anatolij Dvurečenskij. Loomis–sikorski theorem for σ-complete MV-algebras and ℓ-groups. *Journal of the Australian Mathematical Society*, 68(2):261–277, 2000.

[10] Anatolij Dvurečenskij. Loomis–sikorski theorem for monotone σ-complete effect algebras. *Journal of the Australian Mathematical Society*, 79(3):305–318, 2005.

[11] Arthur Fine. Hidden variables, joint probability, and the Bell inequalities. *Physical Review Letters*, 48(5):291–295, 1982.

[12] D. H. Fremlin. *Measure Theory, Vol. 3: Measure Algebras.* Torres Fremlin, Colchester, 2004.

[13] Hector Freytes. An equational theory for σ-complete orthomodular lattices. *Soft Computing*, 24(14):10257–10264, 2020.

[14] Andrew M. Gleason. Measures on the closed subspaces of a Hilbert space. *Journal of Mathematics and Mechanics*, 6(6):885–893, 1957.

[15] Stanley P. Gudder. *Stochastic Methods in Quantum Mechanics.* North-Holland, New York, 1979.

[16] Yukio-Pegio Gunji, Taichi Haruna, and Daisuke Uragami. Contextuality as a left adjoint: a categorical generation of orthomodular structure. *arXiv preprint arXiv:2603.22353*, 2026.

[17] John Harding. Orthomodular lattices whose MacNeille completions are not orthomodular. *Order*, 8(1):93–103, 1991.

[18] Chris Heunen, Nicolaas P. Landsman, and Bas Spitters. A topos for algebraic quantum theory. *Communications in Mathematical Physics*, 291(1):63–110, 2009. arXiv:0709.4364.

[19] Chris Heunen, Nicolaas P. Landsman, and Bas Spitters. Bohrification. In Hans Halvorson, editor, *Deep Beauty: Understanding the Quantum World through Mathematical Innovation*, pages 271–313. Cambridge University Press, 2011. arXiv:0909.3468.

[20] Alfred Horn and Alfred Tarski. Measures in Boolean algebras. *Transactions of the American Mathematical Society*, 64(3):467–497, 1948.

[21] Gudrun Kalmbach. *Orthomodular Lattices.* Academic Press, 1983.

[22] Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17(1):59–87, 1967.

[23] A. N. Kolmogorov. *Foundations of the Theory of Probability.* Chelsea, 1933.

[24] Patrick S. Mahon. Distributivity and the commensurability of empirical adequacy and realism. Preprint, 2026.

[25] Patrick S. Mahon. When does observational coherence determine probability? Preprint, 2026.

[26] Joseph McDonald and Katalin Bimbó. A Stone-type duality for orthomodular lattices. *Mathematical Logic Quarterly*, 2023. arXiv:2208.07430.

[27] V. Müller. Jauch–piron states on concrete quantum logics. *International Journal of Theoretical Physics*, 32(3):433–442, 1993.

[28] Mirko Navara. Regularity and σ-additivity of states on quantum logics. *Proceedings of the American Mathematical Society*, 115(2):427–429, 1992.

[29] Itamar Pitowsky. *Quantum Probability — Quantum Logic*, volume 321 of *Lecture Notes in Physics*. Springer, Berlin, 1989.

[30] Pavel Pták. Exotic logics. *Colloquium Mathematicum*, 54(1):1–7, 1987.

[31] Pavel Pták and Sylvia Pulmannová. A measure-theoretic characterization of Boolean algebras among orthomodular lattices. *Commentationes Mathematicae Universitatis Carolinae*, 35(1):205–208, 1994.

[32] K. P. S. Bhaskara Rao and M. Bhaskara Rao. *Theory of Charges: A Study of Finitely Additive Measures*. Academic Press, New York, 1983.

[33] Alex Simpson. Measure, randomness and sublocales. *Annals of Pure and Applied Logic*, 163(11):1642–1659, 2012.

[34] Josef Tkadlec. Boolean orthoposets — concreteness and orthocompleteness. *Mathematica Bohemica*, 119(2):123–128, 1994.

[35] Veeravalli S. Varadarajan. *Geometry of Quantum Theory*. Springer, New York, 2nd edition, 1985.

[36] Steven Vickers. A localic theory of lower and upper integrals. *Mathematical Logic Quarterly*, 54(1):109–123, 2008.